

Lab 3: Architectural Pattern Selection & Justification

Problem Statement #58 — Digital Art Commission & Watermarking Portal

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Architecture Selection: We chose the Microservices Architecture for the Digital Art Commission & Watermarking Portal System.

Architectural Choice

The platform is decomposed into four independently deployable backend services — Draft Review & Watermarking, Milestone Payment & Asset Delivery, Commission Tracking, and Reputation & Review — each exposed to a shared Client Portal UI through a dedicated provided interface, and the Payment & Delivery service integrates with the external Payment Gateway through a required interface.

Two Reasons for This Choice

- Independent scaling for workloads with very different resource profiles: FR-001's dynamic watermark rendering is CPU/GPU-intensive and its load spikes whenever artists batch-upload drafts, whereas the Milestone Payment & Asset Delivery service (FR-002) handles comparatively few but high-value transactional calls. A microservices split lets the Draft Review & Watermarking service scale horizontally during upload surges without over-provisioning the payment path, which a single monolithic layer could not do independently.
- Fault isolation between unrelated subsystems: NFR-002 requires the watermarking pipeline and payment-gating subsystem to each independently maintain 99.5%+ uptime with their own retry/fallback logic (failed watermark jobs retry up to 3 times before a manual-review queue). Separating these into distinct services means a degraded or retrying Watermarking service cannot block or slow down Commission Tracking (FR-004) or Reputation submissions (FR-005), which is difficult to guarantee inside a single tightly-coupled layered application.

Security Advantage

Isolating the Milestone Payment & Asset Delivery service as its own component confines all payment-gateway credentials and final high-resolution asset access to a single, narrowly-scoped service. The Client Portal never talks to the Payment Gateway directly and never touches the unwatermarked source file; it only receives a short-lived signed download URL (NFR-001, 60-minute expiry) generated by this one service. This satisfies FR-002's pay-to-unlock model and shrinks the attack surface, since a compromise of the Draft & Watermarking, Commission Tracking, or Reputation services would not expose payment credentials or unpublished high-res artwork.

Performance Benefit

Because each service is deployed and scaled independently, the Draft Review & Watermarking service can be scaled out on demand to keep watermark generation responsive during upload bursts, without that scaling affecting the lighter-weight Commission Tracking or Reputation services. This directly supports the acceptance criteria in FR-004 (status updates visible within 2 seconds) and FR-003 (revision requests logged within 1 minute) by ensuring those lightweight, latency-sensitive services are never starved of resources by the much heavier watermarking workload.

Digital Art Commission & Watermarking Portal — Component Diagram

Microservices Architecture

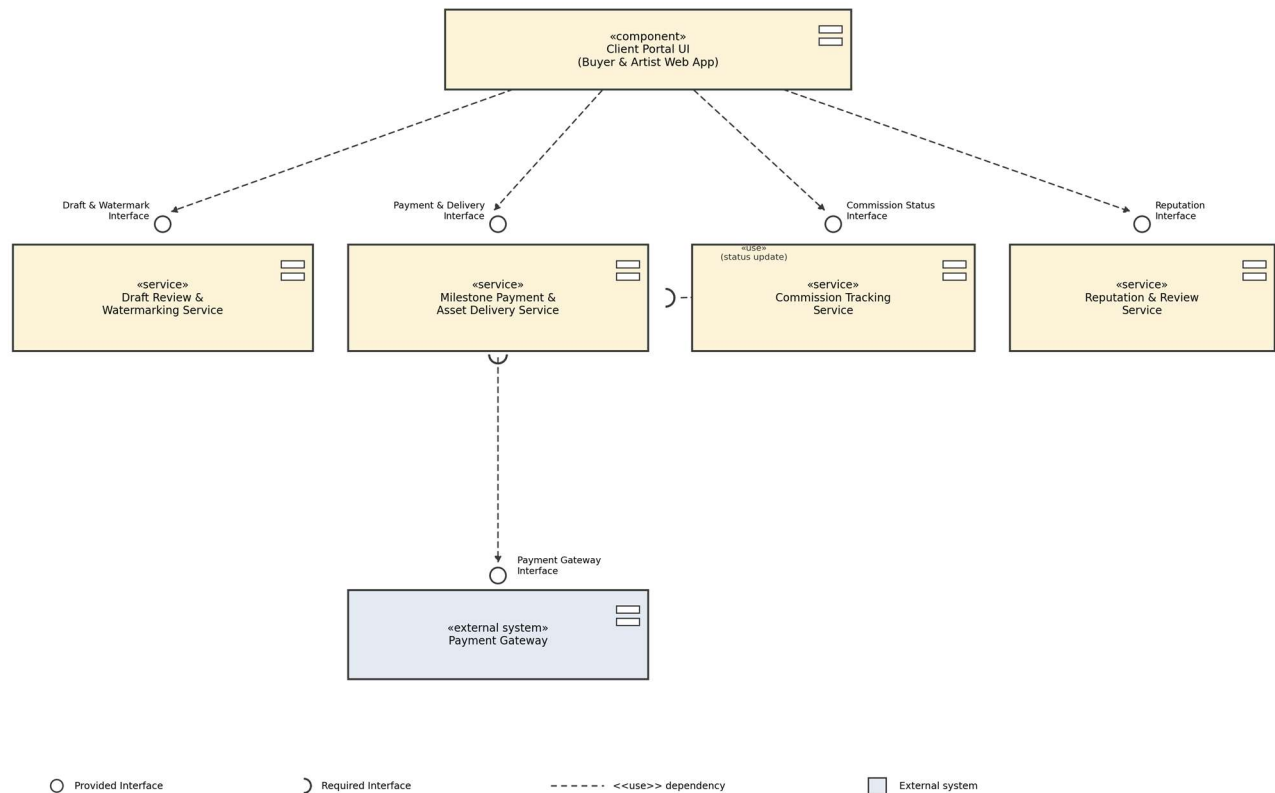


Diagram Reference: See Component Diagram (PNG) — 5 components (Client Portal UI, Draft Review & Watermarking Service, Milestone Payment & Asset Delivery Service, Commission Tracking Service, Reputation & Review Service) plus the external Payment Gateway, with 5 interfaces shown using provided/required UML notation.